P vs NP Proof Attempts:
Crux-First and Steelman Lean Notes on a Weakness-Based Route

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Abstract

This note records the current Lean work around a claimed proof of $P \neq NP$ via "quantale weakness and geometric complexity." The project now proceeds on two tracks in parallel.

The *blocker* track asks where the argument must fail unless an additional theorem is supplied. The *optimistic* track assumes the high-level strategy may work and builds the missing background theory from the ground up so that any genuine repair can plug into a machine-checked scaffold.

The current Lean work proves concrete obstructions. First, a decoder that is merely "local on a bounded-degree neighborhood of radius $\Theta(\log m)$" is *not* thereby polylogarithmic in size: the visible neighborhood already has polynomial size in m. Second, if one does not prove an additional compression theorem, then the class of all local rules on that neighborhood is doubly exponential in the neighborhood size, so it cannot be uniformly represented by a polylogarithmic code budget.

Before the switching story even starts, the sampler already carries another burden. The manuscript fixes the VV width at $k = c_1 \log m$, but the cited $\Omega(1/m)$ uniqueness probability comes from averaging over many scales of k. The new Lean obstruction packages the fixed-width variance bound: if the retained-solution count has mean $\mu > 1$ and variance at most μ, then exact isolation occurs on at most a $\mu/(\mu-1)^2$ fraction of hash choices. So fixed logarithmic width can only isolate often when the pre-hash solution set is already essentially polynomial in size.

And even the manuscript's repeated *pairwise-independent columns* language cannot shoulder the hashing argument by itself. A separate Lean countermodel shows that pairwise-independent columns do not imply universal hashing: one can have pairwise-uniform columns together with a deterministic three-column linear dependency, so a nonzero difference hashes to zero with probability 1.

Nor can the manuscript's δ-biased right-hand side rescue the hash moments by itself. Another Lean module shows that once a candidate's label map is genuinely uniform over the hash-label space, its average hit mass against *any* finite probability weight on labels is exactly the uniform value $1/|L|$; and if a pair of candidates has jointly uniform labels, then the weighted average joint-hit mass is exactly $1/|L|^2$. So any useful effect of the δ-biased right-hand side must come from failure of the underlying label maps to be genuinely uniform, not from the bias itself.

And even exact pre-conditioning uniformity is not enough by itself. A further Lean countermodel shows that conditioning on a label-dependent event can collapse a perfectly uniform label map to a point mass. So any appeal to pre-conditioning 2-universality or uniform labels must be supplemented by a separate theorem about what survives the manuscript's later conditioning on uniqueness.

Third, even one natural repair to the calibration step has a sharp failure mode: if an involution flips the target bit while preserving the retained local features, then every classifier using only those invariant features has exactly $1/2$ average accuracy. So one cannot repair the sign-flip calibration bug merely by dropping the involution-sensitive part of the local input and hoping marginalization will preserve a domination claim.

Even before that charitable repair, the manuscript's exact local input formula already has a hard consistency problem: it defines the post-switch input as $u = (z, a_i, b)$ while the involution T_i toggles b by a_i. So full-input invariance can hold only on the zero-column case $a_i = 0$.

The new bridge module sharpens that fork: on the exact same T_i-action, the reduced projection (z, a_i) is invariant and therefore carries zero symmetry-respecting signed signal, while any advantage obtained by also retaining b is bounded by the mass on nonzero VV columns, since those are exactly the points where b resolves an involution pair.

Sharper still, the same involution hypotheses force every retained feature fiber to contain exactly half target-1 and half target-0 points. So even soft scores or conditional-mean estimators on those invariant features have zero margin on every feature value.

And this is not just a uniform-counting artifact: any involution-invariant finite weighting still assigns equal total mass to the target-1 and target-0 parts of every retained feature fiber. So non-uniform but symmetry-respecting sampling does not restore a usable margin.

In fact, any soft score computed only from those involution-invariant retained features has zero weighted signed correlation with the target. So even a richer invariant scoring statistic carries no usable directional signal before genuine symmetry breaking is introduced.

More sharply, even when the weighting is not exactly involution-invariant, the entire invariant-score signal is still carried only by the antisymmetric part of that weighting. So this rescue route now needs a real theorem that substantial useful mass lives in the weight asymmetry itself.

Likewise, if one augments the invariant local view by a side channel, then any advantage gained by classifying on the pair (u, v) is still bounded by the total mass where the side channel actually separates an orbit point from its involution partner.

Equivalently, any claimed domination gap over chance now forces a matching amount of orbit-resolving mass. So one cannot even state a serious advantage theorem without simultaneously proving a serious resolution theorem.

More generally, keeping a controlled amount of non-invariant information only helps on the slice where it really breaks the symmetry. Any residual slice on which the retained features are still involution-invariant remains exactly at chance under symmetry-respecting weights.

Quantitatively, any excess over chance is bounded by the weight of the symmetry-broken region. So a proof that keeps only a little non-invariant information must also show that almost all relevant mass actually lies outside the unresolved symmetric slice.

Fourth, the natural "short global program" rescue also fails on its own: if the switched input really has only $n = O(\log m)$ visible bits, then an arbitrary local rule on those bits can be hard-wired by a truth table of size $2^n = \mathrm{poly}(m)$. So polynomial-size decoders do not by themselves force any small subclass inside the full local-rule space.

Fifth, the most creative metagraph-style rescue faces its own exactness barrier: if a quotient state must capture the full interface behavior of a k-bit local gadget, then there are 2^{k2^k} such boundary transducers. At binary-log width this already forces at least linear code size in m, so a small exact metagraph/message quotient cannot appear by default.

Sixth, even the tree-message route does not follow from tree-likeness alone: a fixed perfect binary tree architecture of depth n already realizes every Boolean rule on n visible bits, just by changing its leaf labels.

The v13 crux ledger adds two smaller but sharper obligations. A $P = NP$ upper-bound decoder that uses an arbitrary SAT-search output cannot recover a hidden message unless every satisfying witness has the same message readout. The current Lean interface packages a SAT-search output as an arbitrary satisfying witness and proves that correctness for every valid SAT-search output is exactly the fixed-message readout theorem; in the satisfiable case, this is equivalent to single-message readout constancy. Bare satisfiability plus a readout map is not enough: the two-point Boolean model already has two valid SAT-search outputs with different readouts, so no message can be correct for all valid outputs. Separately, unconditional one-coordinate half-success estimates do not imply a product small-success bound. Two identical half-mass events have half-mass intersection, not quarter-mass intersection. Thus the product theorem needs a genuinely sequential conditional hypothesis over admissible histories. The same Lean file now records this repair target positively on the four-point Boolean square: a first-step half-success bound plus a second-step conditional half-success bound implies the two-step product bound. The correlated model fails exactly the second, conditional premise. The separate finite switching module generalizes this to arbitrary finite histories: if every successive event halves the current admissible history class, then the whole tower has the expected product bound. What remains is not product arithmetic, but an instantiation theorem for the actual switched histories. The same generic layer also proves the corresponding failure principle: one failed prefix half-step blocks sequential admissibility for the entire remaining history. The newest monotonicity lemmas also rule out a spurious edge case: if the current history class is already empty, the next prefix half-step is automatic. Hence real blockers must occur at positive-mass conditioning prefixes. The generic layer now also records the global contrapositive: if the final intersection is too large for the claimed tower, then no sequential half-admissibility proof can exist at all. The newest localization theorem strengthens this from a global audit to a specific missing premise: any such product-bound violation yields a decomposition `pre ++ E :: suffix` whose accumulated prefix fails the conditional half-step. The v13 layer lifts the same statement from event lists to switched-coordinate lists, so a failed product count identifies an exact switched coordinate whose induced success event does not halve the current success history. The v13 layer also specializes the product check to abstract atom ledgers, semantic concrete-cell certificates, and ordered fixed-field certificates. Thus a proposed switched list must first pass the tower-product count check before any finer cell structure can rescue it. The v13 instantiation layer now sharpens this further. Lean first proves a useful negative interface fact: an abstract atom ledger with arbitrary per-cell counts is equivalent to the generic prefix half-step, because a successful half-step can always be represented by one summary cell. It then proves the same collapse for an existential "concrete" cell-map certificate: the constant cell map is enough whenever the prefix half-step already holds. The latest Lean theorem lifts this prefix collapse through the whole ordered switching list: both the abstract atom-ledger switching interface and the existential concrete-cell switching interface are equivalent to generic sequential half-admissibility of the induced success events. So a real manuscript instantiation cannot be an existential over cell maps. It must instead name the actual finite history field before each switched coordinate and prove cellwise half-success for that fixed field. The newest atom-ledger structure theorem clarifies the role of the concrete cell map: once a finite `cellOf` is supplied, the prefix and next-success decomposition equations are automatic finite partition identities. Thus decomposition is not an independent escape hatch; the remaining mathematical burden is the cellwise half-success inequality for the actual cells. The fixed-field interface records this obligation: the supplied field's cells must decompose both the current success prefix and the prefix plus the next success event, and the next success event must occupy at most half of each cell. This has now been rewritten in Lean in the more diagnostic form `v13FieldPrefixInstantiation_iff_success_le_failure`: after the finite cell is fixed, a certificate exists exactly when every cell has at least as many next-failure prefix points as next-success prefix points. The supporting theorem `v13ConcretePrefixCount_eq_success_add_failure` proves that this is not a new assumption but the exact split of each cell's prefix mass into success and failure parts. This fixed-field condition is strictly stronger than the global prefix half-step: on the Boolean square, first-coordinate success is globally half-admissible from the empty history, but the fixed field that reveals that same first coordinate has a 'true' cell where success has conditional mass 1, so no fixed-field certificate exists. Lean now also lifts this fixed-field premise to an ordered fielded switching list: each step carries its own supplied field, and the full fixed-field instantiation is equivalent to the recursive list of these cellwise half-success inequalities. Thus the fixed-field decomposition equalities are no longer open proof obligations; the remaining fielded burden is exactly the half-success ledger over the named cells. A full list of fixed-field certificates implies the generic sequential half-admissibility and hence the finite tower product bound. Conversely, a single failed fixed-field prefix certificate blocks the whole remaining fielded suffix, and the same Boolean first-coordinate one-step model instantiates that blocker. The newest fixed-field theorem abstracts the reason: any positive-mass cell on which the next success event is total violates the half-success inequality. In particular, if the supplied field reveals the next success event itself, then a fixed-field certificate can exist only when that next success prefix has zero mass. The latest necessary-condition theorem gives the same fact in positive form: under any valid fixed-field certificate, every cell containing a successful point must also contain an unsuccessful prefix point. A singleton successful atom is therefore already fatal. The newest strengthening generalizes the success-revealing-field blocker: `V13HistoryFieldDeterminesSuccessOn` says that every field cell is homogeneous for the next success event on the current history support, and the corresponding positive-success blocker proves that any such field is certifiable only when the next-success prefix has zero mass. The older global predicate `V13HistoryFieldDeterminesSuccess` is now just a special case. Thus a proposed v13 history field cannot already decide the next success bit after conditioning on any positive-mass prefix; it must leave enough same-cell counterexamples to satisfy the success-versus-failure balance. Lean also checks the repair-resistant distinction: on the Boolean square, the second-coordinate field does *not* globally determine first-coordinate success, but after conditioning on the diagonal history event it does determine that success bit, and the positive diagonal success prefix has no fixed-field certificate or same-cell failure matching. The Boolean first-coordinate field now fails by the global special case, not only by an arithmetic calculation on one named cell. The blocker has also been lifted from the head of a fielded suffix to an arbitrary later position: a failed fixed-field prefix certificate after any preceding fielded list blocks the whole original list, with the current history equal to the old history extended by the preceding success events. In the Boolean square, the one-cell field certifies the first half-success cut, but the first-coordinate field is still impossible as the second repeated-coordinate field because it determines the repeated success event on a positive-mass prefix. The newest interface exposes the same burden constructively: `V13FieldFailureMatching` requires, for every supplied field cell, an injection from successful prefix points into unsuccessful prefix points in that same cell. Lean proves this matching interface equivalent to the fixed-field certificate, and the recursive fielded version is equivalent to the whole fixed-field switching instantiation. Thus a proposed repair can now be asked for an explicit same-cell matching at each switched position, not merely a global counting slogan. The matching interface also gives the contrapositive in the form needed for audit: if the accumulated next-success prefix has positive mass and the recursive matching proof exists, then the supplied field at that position cannot determine the next success bit on the accumulated history support. The newest pointwise extraction theorem makes the same obligation more operational: for every successful prefix point, a recursive matching proof yields an actual failed prefix point in the same supplied field cell. Consequently, one successful point whose accumulated-history cell contains only successes already blocks both the matching formulation and the fixed-field certificate. The diagonal Boolean-square example now also fails by this pointwise criterion, not only by count arithmetic. The same v13-specific module also checks the smallest positive-mass failure mode for sequential repetition: if the second switched step repeats the first exact-success event, then the first success prefix has count 2, the repeated-success prefix still has count 2, and the second prefix half-step is false. Thus no v13 switching certificate can exist for this repeated two-step switching claim. The same example also fails the global product-bound audit: two identical half-mass success events still have half-mass intersection, so the required quarter-mass tower bound is false. The product-bound contrapositive now reaches the operational fixed-field matching layer as well: a tower-product violation rules out the recursive same-cell matching proof directly. In particular, even using the non-revealing one-cell field for both repeated Boolean-square cuts cannot instantiate the fixed-field route; the fielded product bound and the recursive matching obligation both fail. The newest localization theorem also turns either a failed recursive matching proof or a tower-product violation into a concrete failed fielded position $pre ++ item :: suffix$, with the accumulated history computed from pre. For the repeated one-cell Boolean-square model, Lean identifies the exact obstruction: after the first success event, the one-cell field's only accumulated-history cell has next-failure count 0 and next-success count 2, so it is the concrete bad cell for the second repeated success test.

Assessment after the v13 switching packets: the formal work does not prove that no conceivable v13-style switching strategy can work, because one residual positive target remains–name the actual manuscript history fields and prove their recursive cellwise half-success bounds. But it does close the main bookkeeping escape hatches. Abstract ledgers and existential concrete cell maps add no strength beyond generic sequential half-admissibility; decomposition by finite cells is automatic; and fixed fields reduce exactly to the cellwise half-success inequalities, with positive-mass total-success cells and repeated success events blocked by small finite models.

At the same time, the optimistic Lean work now isolates the precise positive interface one would need for the switching lemma to go through: a *small explicitly encoded hypothesis family* whose realized classifier class is bounded by its code budget, together with a perspective-relative notion of accessible input surface inspired by the existing Hyperseed formalization.

These results do not by themselves disprove the whole paper. They do isolate an essential gap: the switching/locality step must be strengthened by a real theorem showing that the specific switched predictors lie in an exceptionally small subclass, not merely in the class of all local functions.

Contents

1 Context

The proof strategy under review is the October 2025 manuscript *"P != NP: A Non-Relativizing Proof via Quantale Weakness and Geometric Complexity."* At a high level, the paper proposes the following chain:

1. Construct a masked random Unique-SAT ensemble.
2. Show that every short polynomial-time decoder can be wrapped into a predictor that is local on a constant fraction of blocks.
3. Use neutrality and sparsification to show local predictors succeed only with exponentially small probability across many independent blocks.
4. Convert that small-success statement into a lower bound on time-bounded Kolmogorov complexity.
5. Contradict the $P = NP$ upper bound obtained from Unique-SAT self-reduction.

The current crux-first effort does not attempt the full chain at once. Instead it attacks the most fragile inference in the middle:

> "Switched predictor is local at logarithmic radius" $\Longrightarrow$ "switched predictor belongs to a polylogarithmic hypothesis class."

That implication is not automatic. It needs either a structural theorem about the switched predictor or a direct circuit/description-length bound. The Lean work below shows why.

2 Current Lean Files

The current Lean modules under `Mettapedia/Computability/PNP/`:

File	Role
`HypothesisClass.lean`	optimistic code-indexed hypothesis families and Hyperseed
`LocalityObstruction.lean`	Log-radius locality is larger than polylog size
`CompressionObstruction.lean`	Arbitrary local rules do not admit short uniform encoding
`FixedWidthIsolationObstruction.lean`	fixed-width hashing isolates only on a small fraction once
`PairwiseSurvivorMoments.lean`	pairwise first and factorial-second moment bounds imply t
`PairwiseCandidateBridge.lean`	candidate-wise 0/1 hits plus pairwise off-diagonal bounds
`PairwiseColumnsObstruction.lean`	pairwise-independent columns still need not form a univer
`RhsBiasIrrelevance.lean`	once label maps are uniform, any finite RHS bias averages
`TwoUniversalRhsIrrelevance.lean`	exact finite 2-universality of the hash family makes RHS b
`ConditioningObstruction.lean`	conditioning can collapse a perfectly uniform label map to
`SymmetrizationObstruction.lean`	unbiased majority does not beat Bayes without a margin
`PostSwitchInputObstruction.lean`	exact $u = (z, a_i, b)$ is T_i-invariant iff $a_i = 0$
`PostSwitchForkObstruction.lean`	exact T_i-fork: invariant (z, a_i) has zero signal, b-side-chan
`VisiblePostSwitchSurface.lean`	isolates the exact visible surface, invariant projection, and
`ExactSwitchedFamily.lean`	isolates the exact switched-family compression target and
`ExactVisibleCompressionObstruction.lean`	if the exact family spans all exact visible rules, no sub-sur
`OrbitNeutralityObstruction.lean`	involution-invariant feature collapse forces exact half accu
`FiberNeutralityObstruction.lean`	invariant feature fibers have exact zero conditional margin
`WeightedFiberNeutralityObstruction.lean`	symmetry-respecting weights still give zero conditional ma
`InvariantScoreObstruction.lean`	invariant soft scores have zero weighted signed signal
`WeightAsymmetryObstruction.lean`	invariant-score signal comes only from antisymmetric weig
`SideChannelResolutionObstruction.lean`	side channels help only on orbit-resolving mass
`ResolutionDemandObstruction.lean`	any domination gap forces matching orbit-resolving mass
`ResidualSymmetryObstruction.lean`	any unresolved symmetric slice still stays at chance
`AsymmetryBudgetObstruction.lean`	advantage is bounded by the symmetry-broken mass
`ShortProgramObstruction.lean`	polynomial-size programs already realize all log-width loca
`MetagraphObstruction.lean`	exact metagraph/interface quotients have state explosion
`TreeMessageObstruction.lean`	fixed tree architectures already realize all local rules
`PNPv13CruxLedger.lean`	v13 decoder readout constancy and product-success finite
`PNPv13CruxLedgerRegression.lean`	regression checks for the v13 crux ledger
`FiniteSwitchingAdmissibility.lean`	generic finite history admissibility, monotone history coun
`FiniteSwitchingAdmissibilityRegression.lean`	regression checks for the generic switching interface
`PNPv13SwitchingInstantiation.lean`	v13 summary/concrete equivalences, fixed history-field an
`PNPv13SwitchingInstantiationRegression.lean`	regression checks for the v13 switching-instantiation interf

These compile in the current `mettapedia` workspace via `lake env lean` (with selected files also built to `.olean` artifacts).

3 Two Parallel Programs

3.1 The Blocker Program

The blocker program is adversarial. It asks:

1. Is the visible radius already too large for any naive `poly(log m)` claim?
2. Is the full class of local rules too large for a short description?
3. Can the manuscript's fixed $k = c_1 \log m$ VV layer really achieve $\Omega(1/m)$ uniqueness probability on large solution sets, or does fixed-width isolation already collapse once the expected survivor count is bigger than 1?
4. Is the manuscript's weaker *pairwise-independent columns* language alone enough to justify universal hashing, or can higher-order linear dependencies still make a nonzero difference hash to zero deterministically?
5. If the hash labels are already uniform or pairwise-uniform, can the δ-biased right-hand side still change the first and pair hit moments at all?
6. Even if the pre-conditioning label map is uniform, can later conditioning on uniqueness still collapse the local label distribution completely?
7. Does majority symmetrization fail whenever the local conditional mean has zero margin?
8. Is the exact post-switch input $u = (z, a_i, b)$ really preserved by the paper's involution, or only in the zero-column case $a_i = 0$?
9. Once that exact fork is made explicit, does the charitable repair reduce either to zero invariant-feature signal on (z, a_i) or to advantage bounded by the mass on nonzero VV columns?
10. Does the obvious "drop the bad coordinate" calibration patch collapse any feature-only classifier back to exact neutrality?
11. Does the softer "estimate the conditional mean on invariant features" repair still have exact zero margin on every retained feature fiber?
12. Does non-uniform but involution-symmetric weighting avoid that zero-margin collapse?
13. Can an arbitrary invariant soft score recover directional signal, or does it also have zero signed correlation with the target?
14. If the weighting is not exactly symmetric, does invariant-score signal come only from the antisymmetric part of that weighting?
15. If one keeps a non-invariant side channel, is total advantage still bounded by the mass where that side channel actually resolves involution pairs?
16. Conversely, does any claimed domination advantage force a comparably large orbit-resolving mass for that side channel?
17. If one keeps some controlled non-invariant information, do residual unresolved slices still remain exact-chance unless their mass is proved negligible?

18. Can a small symmetry-breaking slice still yield a large global advantage, or is the excess over chance quantitatively bounded by the symmetry-broken mass?
19. Does "short global decoder" alone still allow the full local-rule class at log-width?
20. Does an exact finite-state metagraph/message quotient on a log-width interface already require too many states?
21. Does tree-likeness alone force a small exact message/interpreter class?
22. Does arbitrary SAT-search output recover a unique hidden message without a theorem that all satisfying witnesses have the same readout?
23. Do unconditional half-success marginals imply a product small-success bound without sequential conditional admissibility?

The current Lean answer to all twenty-three is "yes." Therefore the published proof cannot be accepted *as written* unless it supplies additional structural theorems.

3.2 The Optimistic Program

The optimistic program asks what would be enough to save the route. The current working answer is:

locality alone is not enough, but *locality plus explicit compressibility* may be.

Concretely, the switching theorem would need to produce not merely a local rule on a radius-$O(\log m)$ neighborhood, but a family of predictors realized by a code space of size at most $\mathrm{polylog}(m)$ bits. In that case the realized hypothesis class is small, and standard finite-class learning bounds become plausible again.

The new module `HypothesisClass.lean` formalizes exactly this interface:

1. an *encoded family* is a decoder from a finite code space into classifiers;
2. the realized class has cardinality at most the size of the code space;
3. therefore an s-bit family realizes at most 2^s classifiers;
4. no such family can surject onto the full local-rule class once $s < 2^n$.

This is the correct positive abstraction: if this switching argument is salvageable, it must land in one of these small encoded subclasses rather than in the full local-rule space.

3.3 Hyperseed Perspective Surface

The Hyperseed formalization in `Mettapedia/Hyperseed/` already provides a useful observer-relative interface:

- a *perspective* determines what is observable,
- a budget determines what is affordable,
- a guard determines what is admitted.

The optimistic PNP background layer reuses this by treating a decoder's *available input surface* as a subtype of the Hyperseed `availableRegion`. This does not solve switching. It does provide a clean ground-up semantics for what a bounded observer can even use as input, without baking in a bespoke notion of locality for every later argument.

4 Optimistic Background Theorem

The basic positive theorem is elementary but indispensable:

Theorem 4.1 (Code budget bounds realized hypothesis class)**.** *Let H be a family of Boolean classifiers realized by decoding an s-bit code:*

$$\mathrm{decode} : \{0,1\}^s \to \mathcal{F}.$$

Then the realized class satisfies

$$|\mathrm{range}(\mathrm{decode})| \le 2^s.$$

This is formalized in `HypothesisClass.lean` as `BitEncodedClassifierFamily.card_realized_le`.

The contrapositive is equally important:

Theorem 4.2 (Short codes cannot cover the full local-rule class)**.** *If $s < 2^n$, then no s-bit decoder can surject onto the full set of Boolean rules on n visible bits.*

This is formalized as `not_surjective_decode_to_fullLocalRule_of_lt`.

These statements are not the final theorem. They are the right *background theory* for the optimistic route: they tell us exactly what shape of compression theorem the switching step must deliver.

5 Obstruction I: Log-Radius Locality Is Already Polynomial

Suppose a factor graph has bounded degree $d \ge 2$, and a predictor is allowed to inspect a radius-r neighborhood around one target bit. A rough but unavoidable upper/lower size model is

$$\text{visible nodes} \asymp d^r.$$

If $r = \Theta(\log m)$, then this is

$$d^{c \log m} = m^{c \log d},$$

which is polynomial in m, not polylogarithmic in m.

The Lean module `LocalityObstruction.lean` records this in several exact forms.

Theorem 5.1 (Binary-log lower bound)**.** *For every $d, m \in \mathbb{N}$ with $d \ge 2$,*

$$m \le d^{\log_2 m + 1}.$$

This is formalized as the theorem `inputSize_le_neighborhoodSize_at_binaryLogRadius`.

Theorem 5.2 (Power-of-two specialization)**.** *For every $d, n \in \mathbb{N}$ with $d \ge 2$,*

$$2^n \le d^{\log_2(2^n)} = d^n.$$

This appears as `powerOfTwo_le_neighborhoodSize_at_exactBinaryLogRadius`.

Theorem 5.3 (Asymptotic dominance)**.** *Fix $k \in \mathbb{N}$ and real constants $d > 1$, $c > 0$. Then*

$$(\log x)^k = o\Big(d^{c \log x}\Big)$$

as $x \to \infty$.

This is formalized as `polylog_isLittleO_logRadiusNeighborhood`, and the corresponding negative Big-O statement appears as `logRadiusNeighborhood_not_isBigO_polylog`.

5.1 Interpretation

This does *not* say that every local rule requires polynomial-size circuits. It says something weaker but still decisive: locality at logarithmic radius, by itself, does not justify a `poly(log m)` complexity claim. The radius already exposes a polynomial-sized input window.

So if the paper wants a tiny post-switch hypothesis class, it must prove a *compression theorem* about the specific local rules that arise after switching. That is a separate and nontrivial burden.

6 Obstruction Ib: Fixed-Width VV Isolation Does Not Follow from the Cited Lemma

The manuscript's main ensemble fixes

$$k = c_1 \log m$$

in the VV layer. But the cited isolation lemma proving uniqueness probability $\Omega(1/m)$ chooses k *uniformly across all scales* $0, \ldots, m-1$. Those are not the same claim.

The new Lean module `FixedWidthIsolationObstruction.lean` packages the exact finite bound one needs at fixed width.

6.1 Lean theorem

Theorem 6.1 (Variance-budget bound for exact isolation)**.** *Let Y be the number of candidates retained by a random hash choice, on a finite hash space. If*

$$\mathbb{E}[Y] = \mu > 1 \qquad \textit{and} \qquad \mathrm{Var}(Y) \le \mu,$$

then

$$\Pr[Y = 1] \;\le\; \frac{\mu}{(\mu-1)^2}.$$

This is formalized as `exactOneProb_le_of_variance_budget`.

The new companion module `PairwiseSurvivorMoments.lean` formalizes the abstract bridge from pairwise-survivor moments to this variance budget. In that file, a finite model satisfying

$$\mathbb{E}[Y] = \mu \qquad \text{and} \qquad \mathbb{E}[Y(Y-1)] \le \mu^2$$

is shown to obey $\mathrm{Var}(Y) \le \mu$, and therefore the same exact-isolation upper bound. This separates the remaining manuscript obligation cleanly: one still has to prove that the actual fixed-width VV sampler instantiates those pairwise moments.

The newer bridge module `PairwiseCandidateBridge.lean` takes one more step toward the actual VV picture. It proves that if the retained-candidate indicators are 0/1-valued, all have the same marginal hit rate p, and distinct candidates satisfy the pairwise upper bound

$$\mathbb{E}[X_a X_b] \le p^2, \qquad a \ne b,$$

then the total survivor count satisfies the exact abstract interface of `PairwiseSurvivorMomentModel`. Concretely:

- `candidateHitCount_sum` proves the first moment formula;
- `candidateHitCount_factorialSecondMoment_le` proves the factorial second-moment bound;

- `pairwiseCandidateMomentModel` packages these into the survivor-moment model used by the fixed-width isolation theorem.

So the remaining gap has narrowed again. To recover the manuscript's fixed-width uniqueness claim, one now needs an end-to-end theorem that the actual VV hash family satisfies these candidate-wise marginal and pairwise bounds at the fixed logarithmic width used later in the paper.

6.2 Why this hits the manuscript

In the source text, the supporting VV lemma states the classical bound

$$\Pr[|S \cap h^{-1}(0^k)| = 1] \geq c/m$$

only after choosing k uniformly from $0, \ldots, m-1$. But the actual ensemble later fixes $k = c_1 \log m$ and then still says isolation succeeds with $\Omega(1/m)$ probability.

That gap is not cosmetic. Under the standard pairwise-independent VV heuristic, if a formula has N satisfying assignments and the fixed hash width is k, then the retained-count random variable has mean

$$\mu = N/2^k$$

and variance at most μ. The Lean theorem above therefore yields

$$\Pr[\text{exact isolation}] \lesssim \frac{1}{\mu} = \frac{2^k}{N},$$

up to a universal constant once μ is bounded away from 1.

So with fixed $k = O(\log m)$, the sampler can isolate with noticeable probability only when the pre-hash solution set is already $N = O(2^k) = \mathrm{poly}(m)$. The manuscript, however, only cites a much weaker upper bound of the form $N \leq 2^{\alpha m}$, which still permits an exponential solution set. On such instances, fixed logarithmic width gives exponentially small exact-isolation probability rather than $\Omega(1/m)$.

6.3 Consequence

This means the paper now needs a real theorem of one of the following forms:

1. on the relevant random-3CNF mass, the satisfiable formulas entering the VV stage already have only $\mathrm{poly}(m)$ satisfying assignments; or

2. the ensemble does not actually use fixed $k = c_1 \log m$, and the later local-input and finite-alphabet claims must be rewritten accordingly.

Without one of those repairs, the promised block distribution D_m is not supported by the cited VV argument.

7 Obstruction Ic: Pairwise-Independent Columns Do Not Imply Universal Hashing

The ensemble definition samples A from "any 2-universal distribution with pairwise-independent columns," and the later robustness discussion repeatedly foregrounds the *pairwise-independent columns* part. The new Lean module `PairwiseColumnsObstruction.lean` shows that this weaker condition is not enough on its own.

7.1 Lean theorem

Theorem 7.1 (Pairwise columns need not be universal)**.** *There exists a one-bit linear hash family with three columns such that:*

1. *each pair of columns is jointly uniform over* $\mathbb{F}_2^2$*; but*
2. *some nonzero input difference hashes to* 0 *on* every *seed.*

Hence pairwise-independent columns do not by themselves imply the universal-hashing bound required for VV isolation.

This is formalized by the bijectivity theorems `col12_bijective`, `col13_bijective`, and `col23_bijective`, together with `threeColumnHash_badDifference` and `threeColumnHash_not_universal`.

7.2 Why this matters

This does *not* refute a hash family that is genuinely 2-universal. It does refute a common informal shortcut: one cannot replace real universal hashing by mere pairwise independence of the columns. Higher-order linear dependencies can survive that weaker condition and completely destroy the needed collision bound.

So the manuscript now owes a sharper statement here too:

1. either prove that the actual fixed-width VV family is genuinely 2-universal in the sense needed for the candidate-wise moment bridge; or
2. stop appealing to pairwise-independent columns as if that weaker property alone carried the isolation argument.

8 Obstruction Id: RHS Bias Is Irrelevant Once The Label Maps Are Uniform

The manuscript samples the VV right-hand side b from a δ-biased source and suggests that this robustness can be folded into the collision/isolation analysis. The new Lean module `RhsBiasIrrelevance.lean` isolates the exact point where that can and cannot matter.

8.1 Lean theorems

Theorem 8.1 (Uniform labels erase single-hit bias)**.** *Let L be a finite label space with any probability weight w on L, and suppose a finite seed space S splits as $S \cong \{1,\dots,d\} \times L$ with $d > 0$, so that a candidate's label map is exactly the projection onto L. Then*

$$\frac{1}{|S|}\sum_{s\in S} w(f(s)) = \frac{1}{|L|}.$$

This is formalized by `average_mass_of_uniform_labels`.

Theorem 8.2 (Jointly uniform labels erase pair-hit bias)**.** *Under the same hypotheses, if a pair of candidates has jointly uniform labels $S \cong \{1,\dots,d\} \times L \times L$ with $d > 0$, then*

$$\frac{1}{|S|}\sum_{s\in S} \mathbf{1}_{f(s)=g(s)}\, w(f(s)) = \frac{1}{|L|^2}.$$

This is formalized by `average_joint_mass_of_uniform_label_pairs`.

8.2 Why this matters

This does not prove that the manuscript's actual fixed-width VV family has perfectly uniform label maps. It proves the sharper conditional statement: *if* the candidate-wise and pairwise label maps are genuinely uniform, then the choice of a δ-biased right-hand-side distribution is irrelevant to the first and pair-hit moments. The averaged values are exactly the same as under a uniform right-hand side.

So the manuscript cannot get extra collision-control power from δ-bias alone. Any real effect of the right-hand side must come from a failure of the underlying label maps to be uniform in the first place. This narrows the burden again:

1. either prove that the actual fixed-width family's label maps are *not* uniform in a way that helps the needed moment bounds; or

2. stop treating δ-bias itself as if it repaired the missing universal-hashing step.

The new bridge module `TwoUniversalRhsIrrelevance.lean` sharpens this one step further in the manuscript's own language: if the actual A-family is genuinely finite 2-universal, then the weighted single-hit and pair-hit averages are exactly the uniform values for *every* finite right-hand-side distribution w. So on the first two VV moments, the entire burden sits on the A-family, not on the δ-biased choice of b.

But even this sharper bridge is only a *pre-conditioning* statement. The new module `ConditioningObstruction.l` shows that this is the end of what one can get for free: conditioning a perfectly uniform label map on a single label fiber makes the conditional label distribution a point mass. So no argument may pass from pre-conditioning uniformity to the distribution conditioned on uniqueness without an additional theorem controlling that conditioning step.

9 Obstruction II: The Full Local Rule Class Is Enormous

Let n be the number of visible Boolean features after switching. Then the unconstrained class of local predictors is simply

$$\{0,1\}^{\{0,1\}^n},$$

the set of all Boolean functions on n bits. Its size is

$$2^{2^n}.$$

Therefore any *uniform* code space that represents all such rules injectively must have at least 2^n bits.

This is formalized in `CompressionObstruction.lean`.

Theorem 9.1 (Counting bound for local rules)**.** *Let* `LocalRule n` *denote the type of Boolean functions on* `n` *visible bits, and* `BitCode s` *the type of* s*-bit codewords. If*

$$s < 2^n,$$

then there is no injective map

$$\texttt{LocalRule } n \to \texttt{BitCode } s.$$

This is the theorem `no_injective_bitCode_of_lt`.

Theorem 9.2 (Binary-log neighborhood corollary)**.** *If* $d \geq 2$ *and*

$$s < 2^m,$$

then there is no injective map

$$\mathtt{LocalRule}\Big(d^{\log_2 m+1}\Big) \to \mathtt{BitCode}\ s.$$

This is the theorem `no_uniform_code_below_expInput_for_binaryLogNeighborhood`.

Theorem 9.3 (Size lower bound on the full class)**.** *For* $d \geq 2$*,*

$$2^{2^m} \leq \Big|\mathtt{LocalRule}\Big(d^{\log_2 m+1}\Big)\Big|.$$

This is the theorem `expExpInput_le_card_localRule_binaryLogNeighborhood`.

9.1 Interpretation

Again, this does not say that the switched rule class *is* the full local rule class. It says that without a theorem proving otherwise, one cannot pass from "local on n visible bits" to "describable by only polylog(m) bits."

That gap is exactly where the current program remains vulnerable.

10 What These Results Break

The present Lean work targets the following inference schema:

$$\text{short decoder} \Longrightarrow \text{switched local predictor} \Longrightarrow \text{tiny ACC}^0\text{/ERM hypothesis.}$$

The first arrow is the paper's switching claim. The second arrow is where the present obstruction lands.

Remark 10.1. If the switched predictor is only known to be some arbitrary function of a log-radius neighborhood, then:

1. the neighborhood already has polynomial size in the ambient parameter, and
2. the class of all such local rules is far too large to admit a polylogarithmic uniform code.

Therefore a successful proof attempt must supply one of the following:

1. a theorem that the switched predictors belong to a very special subclass (for example, a family with explicit circuit/description bound), or
2. a direct compression algorithm showing that the relevant local rules can be encoded in polylog(m) bits for reasons stronger than locality.

Without such a theorem, the proof's route from locality to tiny hypothesis class is unsupported.

11 Obstruction III: Zero-Margin Symmetrization Cannot Concentrate

The symmetrization route in the paper makes a stronger claim than the two counting obstructions above. It says, roughly, that after taking a polylogarithmic number of promise-preserving sign flips and then taking a majority vote, the resulting surrogate label agrees with the Bayes rule on the local σ-field for all but a negligible fraction of blocks.

There is a direct obstruction here when the conditional surrogate mean lands at exactly $\frac{1}{2}$.

11.1 Lean counting lemma

The Lean module `SymmetrizationObstruction.lean` formalizes the following fact.

Theorem 11.1 (Odd-majority is exactly balanced on unbiased bits)**.** *Let s be odd. Among the 2^s bit-vectors of length s, exactly half have strict majority `true` and half have strict majority `false`.*

The key formal statements are:

- `majority_compl`: complementing all bits flips the majority bit when s is odd;
- `card_majority_true_eq_card_majority_false`: the majority-`true` and majority-`false` classes have equal cardinality;
- `majority_tie_break_not_high_probability`: if a Bayes rule breaks a $p = \frac{1}{2}$ tie toward `true`, then majority on s unbiased samples matches that rule on only half the sample space.

11.2 Application to the Current Symmetrization Wrapper

Now consider the paper's symmetrized surrogate label for a fixed block j and bit i. Take the simplest possible decoder P: the decoder that always outputs 0.

Under a sign flip σ, the paper's own back-mapping rule gives

$$Y_{j,i}^{(r)} = \langle a_{j,i}, \sigma^{(r)} \rangle$$

over $\mathbb{F}_2$, because the decoder output is 0 and the back-map xors in the VV shift.

If $a_{j,i} \neq 0$, then $\langle a_{j,i}, \sigma \rangle$ is an unbiased bit under uniform random σ. Hence the sequence

$$Y_{j,i}^{(1)}, \ldots, Y_{j,i}^{(s)}$$

is a fair sample of surrogate bits, and for odd s the majority vote is exactly balanced.

But the paper's Bayes rule is written with threshold $1[p \geq \frac{1}{2}]$, which at $p = \frac{1}{2}$ chooses the value 1. Therefore, in this zero-margin case, the majority vote matches the stated Bayes rule on only half the seed choices, not with probability $1 - m^{-\Omega(1)}$.

11.3 Consequence

This is materially stronger than the earlier "local does not imply simple" objection. It says:

> The concentration-to-Bayes step in the symmetrization argument is false unless an additional margin theorem is proved.

In other words, one must show that for the relevant blocks and bits,

$$\left|p_{j,i} - \tfrac{1}{2}\right|$$

is bounded below by more than the sampling noise scale. Without such a margin, majority voting cannot recover the Bayes rule uniformly, even for the constant-zero decoder.

12 Obstruction IIIb: The Exact Post-Switch Input Is Not T_i-Invariant

Before moving to a charitable repair, the manuscript's own exact definitions already create a direct problem.

Section 3.2 defines the promise-preserving involution

$$T_i(F^h, A, b) = (F^{\tau_i h}, A, b \oplus a_i),$$

while Section 3.3 defines the post-switch local input

$$u_i(\Phi) = (z(F^h), a_i, b).$$

Appendix A.5 then says this same involution preserves $u = (z, a_i, b)$. The new Lean module `PostSwitchInputObstruction.lean` formalizes the exact contradiction point.

12.1 Lean theorem

Theorem 12.1 (Full-input invariance holds only on zero VV columns)**.** *Let $u = (z, a, b)$ be the exact post-switch local input, and let T act on it by*

$$T(z, a, b) = (z, a, b \oplus a).$$

Then

$$T(u) = u \qquad \Longleftrightarrow \qquad a = 0.$$

Equivalently, if the VV column a is nonzero, then the involution necessarily changes the right-hand side coordinate b.

The key formal statements are:

- `vvToggle_eq_self_iff_zero`: $b \oplus a = b$ iff $a = 0$;
- `tiInputMap_eq_self_iff_zeroColumn`: the exact local input is T_i-invariant iff the VV column is zero;
- `invariantProjection_tiInputMap`: the reduced projection (z, a) is always preserved;
- `tiInputMap_changes_b_of_nonzeroColumn`: if $a \neq 0$, then b really changes.

12.2 Why this matters

This is not a soft complaint about omitted detail. It is a direct incompatibility between the paper's own formulas:

1. the exact local input keeps b,
2. the involution toggles b by a_i,
3. yet the calibration appendix says the involution preserves $u = (z, a_i, b)$.

That can only be true on the special case $a_i = 0$. So the calibration argument cannot be right as written on the full local input.

12.3 Consequence

This forces the proof into a fork:

1. either keep the full input (z, a_i, b), in which case the involution is not feature preserving and a new calibration theorem is needed;
2. or drop the changing coordinate b, in which case one lands in the invariant-feature regime attacked next.

13 Obstruction IIIc: The Exact Manuscript Fork Already Lands in the Symmetry Budget

The previous section exposed the formula-level inconsistency in the manuscript as written. The new Lean module `PostSwitchForkObstruction.lean` shows that the most charitable exact repair is already constrained by the later symmetry-budget files.

13.1 Lean theorems

Theorem 13.1 (The exact T_i-action is an involution and b resolves exactly the nonzero columns)**.** *On the exact local input $u = (z, a, b)$ with*

$$T(z, a, b) = (z, a, b \oplus a),$$

the map T is an involution. Moreover,

$$b(Tu) \neq b(u) \qquad \Longleftrightarrow \qquad a \neq 0.$$

Equivalently, the b-coordinate distinguishes an involution pair exactly on the nonzero-column slice.

Theorem 13.2 (The exact fork)**.** *Let the target bit flip under the exact involution T, and let the weighting be T-invariant. Then:*

1. *every soft score depending only on the reduced invariant projection (z, a) has zero weighted signed correlation with the target;*
2. *every classifier using $((z, a), b)$ has success advantage over chance bounded by the total weight of the nonzero-column slice.*

The key formal statements are:

- `tiInputMap_involutive`: the manuscript's exact local update really is an involution;
- `resolvedBySideChannel_tiInputMap_b_iff_nonzeroColumn`: the b-side channel resolves an orbit pair exactly when $a \neq 0$;
- `weighted_signedScore_sum_eq_zero_on_invariantProjection`: the reduced exact input (z, a) has zero signed signal under symmetry-respecting weights;
- `doubledAdvantage_invariantProjection_with_b_le_nonzeroColumnMass`: any classifier on $((z, a), b)$ pays for its domination gap with nonzero-column mass.

13.2 Why this matters

This turns the manuscript fork into a direct quantitative obligation.

If one drops b, then the surviving exact input is already inside the invariant-feature regime: no symmetry-respecting signed signal remains. If one keeps b, then the proof is no longer allowed to say merely that "a small side channel helps." It must prove that the resulting domination gap is supported by a sufficiently large nonzero-column slice in the actual weighted domain.

So this exact bridge does not merely identify a local inconsistency. It shows that the natural repair lands immediately in the later neutrality and resolution-budget obstructions.

14 Obstruction IV: Involution-Invariant Feature Collapse Forces Exact Neutrality

Once the exact full-input inconsistency is noticed, the most natural charitable repair is to drop the involution-sensitive coordinate b and keep only the involution-fixed part of the local view, say a feature map of the form

$$u' = (z, a_i).$$

This repair looks attractive because it restores feature invariance. The new Lean module `OrbitNeutralityObstruction.lean` shows why this move cannot, by itself, support a domination theorem.

14.1 Lean theorem

Theorem 14.1 (Feature-preserving involutions force exact half accuracy)**.** *Let α be a finite sample space, let $\tau : \alpha \to \alpha$ be an involution, let $u : \alpha \to U$ be a feature map, and let $y : \alpha \to \{0, 1\}$ be the target bit. Assume*

$$u(\tau x) = u(x) \qquad \text{and} \qquad y(\tau x) = 1 - y(x)$$

for every $x \in \alpha$. Then for every classifier $h : U \to \{0, 1\}$,

$$2 \cdot \#\{x : h(u(x)) = y(x)\} = |\alpha|.$$

Equivalently, the average accuracy of $h \circ u$ is exactly $1/2$.

The key formal statements are:

- `correct_under_pair_iff_incorrect`: correctness at x is equivalent to incorrectness at τx;
- `card_correct_eq_card_incorrect`: the correct and incorrect classes have equal cardinality;
- `two_mul_card_correct_eq_card`: exactly half the finite sample space is classified correctly.

14.2 Why this hits the charitable calibration patch

Suppose the repair to the calibration bug is:

1. remove the involution-sensitive coordinate b from the local input;
2. prove neutrality by pairing each instance with its T_i-flipped partner;
3. then learn or evaluate a classifier using only the reduced feature map u'.

If the paired instances really preserve u' and flip the target bit, then the theorem above applies immediately. Every feature-only classifier on u' has exact $1/2$ average accuracy. So this route cannot yield a strict domination statement, a high-probability Bayes recovery statement, or any per-block success advantage above chance.

14.3 Consequence

This is a stronger objection than "the paper needs more detail." It says:

> If the calibration fix works by dropping the coordinate changed by the involution and then classifying from the invariant remainder alone, that fix destroys any chance of beating $1/2$ accuracy on the paired finite domain.

So a viable rescue must do something more subtle than mere feature collapse. For example, it would have to:

1. keep some controlled non-invariant information,
2. switch from hard classification to a softer statistic whose expectation matters,
3. or show that the actual learning/evaluation domain is not the balanced finite orbit space modeled by the theorem above.

15 Obstruction IVb: Invariant Feature Fibers Have Zero Margin

The most natural reply to Obstruction IV is:

> Fine, hard classification on the invariant features fails. But one can still estimate a *soft score* or conditional mean on those features and only use high-margin feature values.

This is the right stronger target. The new Lean module `FiberNeutralityObstruction.lean` blocks that repair under the same involution hypotheses.

15.1 Lean theorem

Theorem 15.1 (Every invariant feature fiber is exactly balanced)**.** *Let α be a finite sample space, let $\tau : \alpha \to \alpha$ be an involution, let $u : \alpha \to U$ be a feature map, and let $y : \alpha \to \{0, 1\}$ be the target bit. Assume*

$$u(\tau x) = u(x) \qquad \text{and} \qquad y(\tau x) = 1 - y(x)$$

for every $x \in \alpha$. Then for every retained feature value $v \in U$,

$$2 \cdot \#\{x : u(x) = v \text{ and } y(x) = 1\} = \#\{x : u(x) = v\}.$$

Equivalently, conditioned on any retained feature value v, the target bit has exact mean $1/2$.

The key formal statements are:

- `featureFiberTrueEquivFalse`: on each fiber $u^{-1}(v)$, the involution pairs the `true` points with the `false` points;
- `card_featureFiberTrue_eq_card_featureFiberFalse`: the two parts of each fiber have equal cardinality;
- `two_mul_card_featureFiberTrue_eq_card_featureFiber`: the `true` points occupy exactly half of each fiber.

15.2 Why this hits the soft-score repair

This is stronger than the hard-classifier obstruction. It says the retained feature map does not merely fail to support a good classifier; it fails to support *any nonzero conditional margin.* On every retained feature value v, the exact local average is already

$$\mathbb{E}[\, y \mid u = v \,] = \tfrac{1}{2}.$$

So if the repair route is:

1. drop the involution-sensitive coordinate,
2. keep only the invariant summary u',
3. estimate a soft score or conditional mean from u',
4. then use only "high-margin" u'-values,

it fails immediately on the balanced finite orbit model. There are no high-margin retained feature values at all.

15.3 Consequence

This narrows the remaining rescue space again. A successful patch now has to do more than "replace hard labels by soft scores." It must either:

1. retain some controlled non-invariant information,
2. show that the actual averaging domain is not fiberwise balanced under the involution,
3. or use a structure theorem stronger than feature-invariant orbit pairing.

16 Obstruction IVc: Symmetry-Respecting Weights Still Have Zero Margin

Another natural reply is:

> The real evaluation or training average is not uniform on the finite orbit space. Once one uses the correct non-uniform weighting, the invariant features may regain a useful soft margin.

The new Lean module `WeightedFiberNeutralityObstruction.lean` blocks that repair whenever the weighting itself respects the same involution symmetry.

16.1 Lean theorem

Theorem 16.1 (Orbit-symmetric weights preserve zero fiber margin)**.** *Let α be a finite sample space, let $\tau : \alpha \to \alpha$ be an involution, let $u : \alpha \to U$ be a retained feature map, let $y : \alpha \to \{0, 1\}$ be the target bit, and let $w : \alpha \to M$ be a weight into any additive commutative monoid. Assume*

$$u(\tau x) = u(x), \qquad y(\tau x) = 1 - y(x), \qquad w(\tau x) = w(x)$$

for every $x \in \alpha$. Then for every retained feature value v,

$$\sum_{x:u(x)=v,\ y(x)=1} w(x) = \sum_{x:u(x)=v,\ y(x)=0} w(x).$$

The key formal statements are:

- `weightedFeatureFiberTrueMass`: the weighted mass of the `true` part of one feature fiber;
- `weightedFeatureFiberFalseMass`: the weighted mass of the `false` part;
- `weightedFeatureFiberTrueMass_eq_weightedFeatureFiberFalseMass`: the exact balance theorem under involution-invariant weights.

16.2 Why this closes the non-uniform averaging reply

This matters because it removes a very common escape hatch. One can no longer say:

> The balanced finite orbit space was just a counting toy model. The real proof uses a more careful empirical or distributional weighting, so the invariant local statistic may still have a nonzero expectation gap.

Not if that weighting respects the same sign-flip involution. The theorem shows that every such weighting still gives exact zero conditional margin on every retained feature fiber. In particular, normalizing an involution-invariant finite weight into a probability distribution still leaves

$$\mathbb{E}[\, y \mid u = v \,] = \tfrac{1}{2}$$

for every retained feature value v with nonzero mass.

16.3 Consequence

So the remaining escape routes are narrower again. To revive a margin on the retained local input, one must now do something stronger than:

1. pass from hard labels to soft scores,
2. pass from uniform counts to non-uniform but symmetry-respecting weights,
3. or restrict attention to invariant-feature fibers alone.

What survives must either break the involution symmetry in the weighting/domain, retain some controlled non-invariant information, or prove a much stronger ensemble-specific structural theorem.

17 Obstruction IVd: Residual Symmetric Slices Still Stay At Chance

The next natural reply is:

> Fine. We keep a little non-invariant information, so the exact sign-flip symmetry is broken. We do not need to resolve every orbit, only enough of them.

This is the right next theorem to ask for. The new Lean module `ResidualSymmetryObstruction.lean` makes the burden precise: every residual slice on which the retained information still fails to distinguish the involution partners remains exact-chance under symmetry-respecting weights.

17.1 Lean theorem

Theorem 17.1 (Any unresolved symmetric slice remains balanced)**.** *Let α be a finite sample space, let $\tau : \alpha \to \alpha$ be an involution, let $p \subseteq \alpha$ be a τ-stable slice, let $u : \alpha \to U$ be the retained local features, let $y : \alpha \to \{0, 1\}$ be the target bit, let $w : \alpha \to M$ be a weight into an additive commutative monoid, and let $h : U \to \{0, 1\}$ be any classifier. Assume on the slice p that*

$$u(\tau x) = u(x), \qquad y(\tau x) = 1 - y(x), \qquad w(\tau x) = w(x).$$

Then the total weighted mass of correctly classified points in p equals the total weighted mass of incorrectly classified points in p.

The key formal statements are:

- `weightedCorrectMass`: total weighted correct mass for a feature-only classifier;
- `weightedIncorrectMass`: total weighted incorrect mass;
- `weightedCorrectMass_eq_weightedIncorrectMass`: the global weighted half-accuracy theorem under exact involution symmetry;
- `weightedCorrectMass_eq_weightedIncorrectMass_on_slice`: the residual-slice version obtained by restricting to a τ-stable subtype.

17.2 Why this hits the "controlled non-invariant info" reply

This is the right quantitative obstruction. A small amount of non-invariant information can only help on the part of the domain where it actually breaks the orbit symmetry. Every remaining slice on which the feature map is still involution-invariant contributes only chance performance.

So the paper can no longer get by with a vague statement of the form:

> We keep a little symmetry-breaking local information, and that should be enough to restore a useful predictor.

To make that route work, one now needs a real theorem that the unresolved symmetric slice carries negligible weight, or else a theorem that the retained information breaks the involution on almost all of the relevant mass.

17.3 Consequence

This does not yet show that *every* controlled symmetry-breaking repair fails. It does show that such a repair only helps to the extent that it resolves most of the mass. What remains must therefore prove one of:

1. the residual symmetric slice has negligible weight,
2. the retained non-invariant information separates almost all involution pairs,
3. or the actual evaluation domain is not well-modeled by a symmetry-respecting weighted slice at all.

18 Obstruction IVe: Advantage Is Bounded By Symmetry-Broken Mass

The next creative rescue move is the quantitative version of the previous one:

> Maybe we only break the symmetry on a relatively small region, but that is enough because the classifier can exploit that region very efficiently and still gain a substantial global advantage.

The new Lean module `AsymmetryBudgetObstruction.lean` blocks that move in the natural finite weighted model.

18.1 Lean theorem

Theorem 18.1 (Asymmetry budget bound)**.** *Let α be a finite sample space with weight $w : \alpha \to \mathbb{N}$. Let $p \subseteq \alpha$ be a τ-stable slice on which the retained feature map is still invariant under an involution τ, the target bit is flipped by τ, and the weights are τ-invariant. Then for every classifier h,*

$$2 \cdot \mathrm{CorrectMass}(h) \;\leq\; \mathrm{TotalMass} + \mathrm{OutsideMass}(p).$$

Equivalently, the global excess over chance is bounded by the total mass outside the unresolved symmetric slice.

The key formal statements are:

- `two_mul_correctSliceMass_eq_sliceMass`: on the unresolved symmetric slice, the correct mass is exactly half of the slice mass;
- `weightedCorrectMass_eq_correctSlice_add_correctOutside`: total correct mass splits into the unresolved slice contribution plus the symmetry-broken contribution;
- `correctOutsideMass_le_outsideMass`: outside the unresolved slice, even perfect classification cannot use more mass than is actually there;
- `two_mul_weightedCorrectMass_le_total_plus_outside`: the final asymmetry-budget inequality.

18.2 Why this hits the last soft escape

This is stronger than saying "the residual slice is still at chance." It says the *entire global advantage* is quantitatively budgeted by the part of the domain where symmetry is actually broken.

So one can no longer say:

> We only need a little symmetry-breaking local information. Perhaps it affects a small region, but that small region could still be enough to drive the whole lower-bound argument.

Not without a theorem that the symmetry-broken region itself carries substantial mass. The Lean bound shows that if the unresolved symmetric slice has mass close to the total, then the global accuracy remains close to 1/2.

18.3 Consequence

At this point, the remaining route is no longer merely "add some asymmetry." It must prove a real mass-transfer statement:

1. either almost all relevant mass leaves the involution-symmetric slice after the repair,
2. or the weighted domain relevant to the proof is not symmetry-respecting in the sense used above,
3. or the real predictor uses stronger ensemble-specific information than the current local symmetry analysis captures.

19 Obstruction IVf: Invariant Soft Scores Have Zero Signed Signal

Another creative move is:

> Perhaps the proof should not be phrased as conditional means on feature fibers at all. Maybe one can compute a richer soft score from the invariant local summary and then feed that score into a more delicate downstream argument.

This is a real strengthening of the repair attempt. The new Lean module `InvariantScoreObstruction.lean` blocks the core signal claim under the same involution symmetry.

19.1 Lean theorem

Theorem 19.1 (Invariant soft scores have zero signed correlation)**.** *Let α be a finite sample space, let $\tau : \alpha \to \alpha$ be an involution, let $u : \alpha \to U$ be a retained feature map, let $y : \alpha \to \{0, 1\}$ be the target bit, let $w : \alpha \to \mathbb{N}$ be a τ-invariant weight, and let* $\mathrm{score} : U \to \mathbb{Z}$ *be any soft score computed only from the retained features. Assume*

$$u(\tau x) = u(x), \qquad y(\tau x) = 1 - y(x), \qquad w(\tau x) = w(x)$$

for every $x \in \alpha$. Writing

$$\mathrm{sgnTarget}(y) = \begin{cases} 1 & y = 1, \\ -1 & y = 0, \end{cases}$$

one has

$$\sum_{x \in \alpha} w(x)\,\mathrm{score}(u(x))\,\mathrm{sgnTarget}(y(x)) = 0.$$

The key formal statements are:

- `targetSign`: the $\{\pm 1\}$-encoding of a Boolean target;
- `targetSign_flip`: flipping the Boolean target negates that signed encoding;
- `weighted_signedScore_sum_eq_zero`: the weighted zero-correlation theorem for any invariant soft score.

19.2 Why this hits the richer invariant-scoring repair

This closes a subtler loophole than the earlier fiber-balance statements. Those showed that the conditional mean on each invariant feature value is exactly $1/2$, and therefore that hard classifiers or direct mean estimators on those features cannot beat chance. The new theorem says more: even before one thresholds or calibrates anything, *the score itself* has zero signed signal against the target.

So one can no longer say:

> We are not really using a classifier on the invariant features. We only need a good enough numerical score from those features, and then a later nonlinear step can extract the useful signal.

Not under the same involution symmetry. Any score depending only on the invariant retained features is paired with its involution partner at equal weight and opposite signed target. The positive and negative contributions cancel exactly.

19.3 Consequence

Taken together with the previous sections, this leaves very little room for an invariant-feature rescue. The remaining route must now prove something outside the invariant scoring model:

1. either the actual score retains genuinely non-invariant information,
2. or the relevant domain/weighting is not symmetry-respecting,
3. or the ensemble has a stronger special structure that invalidates the involution pairing hypotheses on the real predictor family.

20 Obstruction IVg: Invariant-Score Signal Comes Only From Weight Asymmetry

The next and now very natural reply is:

> Fine. Perhaps the retained score inputs stay involution-invariant, but the real training or evaluation weighting is not exactly symmetric. Then an invariant soft score may still pick up a useful signed signal.

This is the last serious invariant-score escape hatch. The new Lean module `WeightAsymmetryObstruction.lean` makes that route exact: any invariant-score signal comes entirely from the antisymmetric part of the weighting.

20.1 Lean theorem

Theorem 20.1 (Only antisymmetric weight mass contributes to invariant-score signal)**.** *Let α be a finite sample space, let $\tau : \alpha \to \alpha$ be an involution, let $u : \alpha \to U$ be a retained feature map, let $y : \alpha \to \{0, 1\}$ be the target bit, let $w : \alpha \to \mathbb{N}$ be an arbitrary weight, and let* $\mathrm{score} : U \to \mathbb{Z}$ *be any soft score on the retained features. Assume*

$$u(\tau x) = u(x), \qquad y(\tau x) = 1 - y(x)$$

for every $x \in \alpha$. Then

$$2 \sum_{x\in\alpha} w(x)\, \mathrm{score}(u(x))\, \mathrm{sgnTarget}(y(x)) = \sum_{x\in\alpha} (w(x) - w(\tau x))\, \mathrm{score}(u(x))\, \mathrm{sgnTarget}(y(x)).$$

The key formal statements are:

- `signedScoreContribution`: one point's contribution to the invariant signed-score sum;
- `antisymmetricWeightContribution`: the corresponding contribution of the weight-difference term $w(x) - w(\tau x)$;
- `two_mul_signedScore_sum_eq_antisymmetricWeight_sum`: the exact decomposition identity.

20.2 Why this narrows the weighting rescue

This is stronger than the earlier exact-symmetry obstruction. It says not just that symmetric weights give zero signal, but that *all* invariant-score signal is sourced only by the antisymmetric part of the weighting. The involution-invariant part cancels exactly.

So one can no longer say:

> The true weighting is probably a little asymmetric, and that may be enough.

Not without quantifying that asymmetry. The theorem shows that the proof now needs a real bound of the form:

> the antisymmetric weight budget is large enough on the relevant domain to support the claimed advantage.

20.3 Consequence

This pushes the remaining route into a much narrower quantitative corner. To save invariant local scoring, one must now prove one of:

1. the actual proof-relevant weighting has a substantial antisymmetric component,
2. the score retains genuinely non-invariant information, or
3. the real ensemble/predictor pair is not governed by the involution model above.

21 Obstruction IVh: Side Channels Help Only On Orbit-Resolving Mass

At this point the natural pivot is:

> Fine. We stop asking invariant features alone to do the whole job. We keep a controlled involution-sensitive side channel v, and classify on the pair (u, v).

This is the right next charitable target. The new Lean module `SideChannelResolutionObstruction.lean` packages the earlier symmetry budget results into the exact theorem that this move needs to face.

21.1 Lean theorem

Theorem 21.1 (Side-channel advantage is bounded by orbit-resolving mass)**.** *Let α be a finite sample space, let $\tau : \alpha \to \alpha$ be an involution, let $u : \alpha \to U$ be a retained invariant feature map, let $v : \alpha \to V$ be an arbitrary side channel, let $y : \alpha \to \{0, 1\}$ be the target bit, let $w : \alpha \to \mathbb{N}$ be a τ-invariant weight, and let $h : U \times V \to \{0, 1\}$ be any classifier. Assume*

$$u(\tau x) = u(x), \qquad y(\tau x) = 1 - y(x), \qquad w(\tau x) = w(x)$$

for every $x \in \alpha$. Define the resolved mass *to be the total weight of the points for which the side channel separates the involution pair:*

$$\text{ResolvedMass}(v) = \sum_{x:v(\tau x)\neq v(x)} w(x).$$

Then

$$2 \cdot \text{CorrectMass}(h \circ (u, v)) \ \leq \ \text{TotalMass} + \text{ResolvedMass}(v).$$

Equivalently, the entire global excess over chance is bounded by the weight of the domain where the side channel actually resolves the orbit pairing.

The key formal statements are:

- `unresolvedBySideChannel`: the slice on which $v(\tau x) = v(x)$;
- `resolvedMass`: the weight outside that unresolved slice;
- `two_mul_weightedCorrectMass_pair_le_total_plus_resolvedMass`: the final resolution-budget inequality for classifiers on (u, v).

21.2 Why this hits the controlled-side-channel rescue

This is stronger than the earlier residual-symmetry prose, because it says exactly what the side channel must buy. Keeping some involution-sensitive information does *not* by itself create a large advantage. The benefit is quantitatively capped by the mass where that side channel really separates involution partners.

So one can no longer say:

> We only need a small side channel. Once it is present, a clever classifier on (u, v) may still recover a substantial domination advantage.

Not without a theorem that the side channel resolves involution pairs on a substantial fraction of the relevant mass. The unresolved slice remains chance, and the total excess over chance is budgeted by what lies outside it.

21.3 Consequence

This leaves the controlled-side-channel route in a much sharper form. To save it, one must now prove one of:

1. the retained side channel separates involution pairs on most of the proof-relevant mass,
2. the weighting itself has a large antisymmetric component that can be exploited,
3. or the real predictor is not well-modeled by this involution-paired finite setup.

22 Obstruction IVi: Any Domination Gap Forces Matching Resolution

The previous section bounded success by resolved mass. The converse reading is now just as important:

> if the proof claims a real domination gap over chance from a retained side channel, that claim already commits it to a quantitatively comparable orbit-resolution theorem.

The new Lean module `ResolutionDemandObstruction.lean` packages that commitment as a direct corollary.

22.1 Lean theorem

Theorem 22.1 (Doubled advantage is bounded by resolved mass)**.** *For a weighted classifier h, define the* doubled advantage

$$\mathrm{DoubledAdv}(h) = 2 \cdot \mathrm{CorrectMass}(h) - \mathrm{TotalMass}.$$

Under the same involution hypotheses as in Obstruction IVh, one has

$$\mathrm{DoubledAdv}(h \circ (u, v)) \leq \mathrm{ResolvedMass}(v).$$

The key formal statements are:

- `doubledAdvantage`: twice the weighted success gap over chance;
- `doubledAdvantage_pair_le_resolvedMass`: the final converse bound.

22.2 Why this matters

This is the strongest current version of the side-channel blocker. It no longer says only that resolved mass is *sufficient* for advantage. It says a claimed advantage is itself evidence that substantial resolution must already be present.

So one can no longer say:

> Perhaps the predictor gets a modest but important edge from a very small side channel, and only later amplifies that edge through clever structure.

Not unless that modest edge is matched by actual orbit-resolving mass. The theorem makes the burden explicit: any nontrivial domination statement already entails a nontrivial local resolution statement.

22.3 Consequence

This is close to the form needed to challenge the route on its own terms. To save the current style of proof, one now needs a theorem of the form:

> the switched predictor's retained side channel resolves involution pairs on enough mass to support the claimed lower-bound-driving advantage.

Without that, the claimed domination gap has no quantitative room to exist.

23 Obstruction V: Short Programs Still Cover the Full Log-Width Local Rule Class

Another natural reply is:

> The switched predictor comes from a short global program, so even if arbitrary local rules are huge, only a much smaller subclass can actually arise from polynomial-time decoders.

This is not true without an additional restriction beyond mere global shortness.

23.1 Lean theorem

The new module `ShortProgramObstruction.lean` formalizes the basic truth-table fact.

Theorem 23.1 (Truth tables realize all local rules)**.** *For every n, there is a surjective decoder*

$$\{0,1\}^{2^n} \twoheadrightarrow \{\, f : \{0,1\}^n \to \{0,1\} \,\}.$$

Equivalently, every Boolean rule on n visible bits can be hard-wired by a truth table of length 2^n.

The key formal statements are:

- `truthTableDecode_surjective`: every local rule is realized by a 2^n-bit truth table;
- `truthTableFamily_realizes_all`: the realized class of the truth-table family is the full local-rule space;
- `card_realized_truthTableFamily`: this realized class has exactly cardinality 2^{2^n}.

23.2 Binary-log consequence

At binary-log width the truth-table size is already linear in the ambient parameter:

$$2^{\lfloor \log_2 m \rfloor + 1} \leq 2m \qquad (m \neq 0).$$

This is formalized as `truthTableBits_binaryLogWidth_le_twoMul`.

So if the switched interface really has only $n = \lfloor \log_2 m \rfloor + 1$ visible bits, then a merely linear-size code budget already suffices to realize the *entire* local-rule class on that interface.

23.3 Consequence

This directly blocks the short-program rescue in its naive form:

> "The global decoder is polynomial-size" does not imply "the induced local rule lies in a tiny subclass."

On the contrary, polynomial-size descriptions are already large enough to hard-wire every Boolean rule on $O(\log m)$ visible bits. So if one wants a true small subclass theorem, it must use something much stronger than global shortness alone: symmetry, algebraic normal form, a tree interpreter with bounded message alphabet, or some other genuine structural restriction.

24 Obstruction VI: Exact Metagraph Quotients Have Interface-State Explosion

The strongest remaining charitable rescue is roughly this:

> The switched neighborhood first collapses to a canonical metagraph or message-passing quotient, and the witness bit is computed by a fixed interpreter on that quotient.

This is a serious idea. It is also constrained by a generic exactness barrier.

24.1 Lean theorem

The new module `MetagraphObstruction.lean` models a quotient state by its exact interface behavior on k visible boundary bits. The semantic object is then a boundary transducer

$$(\{0,1\}^k \to \{0,1\}^k).$$

Theorem 24.1 (Boundary transducers are exponentially numerous)**.** *There are exactly*

$$2^{k2^k}$$

boundary transducers on k visible bits.

The key formal statements are:

- `card_boundaryMap`: exact k-bit interface semantics have cardinality 2^{k2^k};
- `no_surjective_bitCode_to_boundaryMap_of_lt`: no s-bit code can cover the full interface-semantic space when $s < k2^k$;
- `no_exact_metagraphQuotient_below_inputSize_binaryLogWidth`: at $k = \lfloor \log_2 m \rfloor + 1$, there is no exact quotient with fewer than m code bits.

24.2 Why this matters

If a metagraph/message quotient is meant to be *exact*, then it must preserve enough state to reconstruct the gadget's interface behavior under gluing. The theorem above says that, in the absence of an additional structural restriction, the exact interface-semantic space is already far too large for a tiny finite-state algebra.

This does not yet show that the current ensemble realizes the full transducer class. What it does show is that a small exact quotient theorem cannot follow from tree-likeness or locality alone. To survive, the metagraph route needs a new theorem that the actual switched/VV gadgets lie in a drastically smaller semantic subclass.

24.3 Consequence

So the metagraph rescue is now narrowed to something like:

> not "there is an exact small quotient of all local interfaces," but "the specific random masked+isolated interfaces collapse to a much smaller algebra for ensemble-specific reasons."

That is a legitimate possible theorem. It is also substantially stronger than anything currently written in the paper, and it is the next point where a decisive break may occur.

25 Obstruction VII: Tree-Likeness Alone Does Not Yield a Small Message Class

Another natural repair is:

> On locally tree-like neighborhoods, the witness bit is computed by a small message-passing or dynamic-program interpreter on the tree.

This too needs a stronger premise than mere tree structure.

25.1 Lean theorem

The new module `TreeMessageObstruction.lean` defines a fixed perfect binary tree architecture of depth n, where internal nodes merely branch on the next input bit and only the leaves carry data.

Theorem 25.1 (Perfect binary trees realize all local rules)**.** *For every Boolean rule*

$$f : \{0,1\}^n \to \{0,1\},$$

there is a depth-n perfect binary decision tree whose evaluation equals f.

The key formal statements are:

- `eval_encodeDecisionTree`: every local rule is exactly realized by a leaf-labeled depth-n perfect tree;
- `evalDecisionTree_surjective`: the semantics map from such trees onto `LocalRule n` is surjective;
- `no_surjective_bitCode_to_decisionTreeSemantics_of_lt`: therefore no s-bit exact code can cover all depth-n tree semantics when $s < 2^n$.

25.2 Why this matters

This is a direct answer to the naive tree-message rescue:

> tree-likeness by itself does *not* imply semantic simplicity.

Even with an extremely rigid tree architecture, the semantic class is already the full Boolean function space on the visible inputs. So if one wants a genuine tree-message theorem, it must use special properties of the actual SAT+VV messages, not just the fact that the underlying graph is a tree.

25.3 Consequence

The surviving charitable tree route is now much narrower:

> not "all tree-like local neighborhoods admit a tiny interpreter," but "the specific masked+isolated tree messages belong to a sharply smaller algebra."

That is again a real possible theorem. It is also much stronger than the current text, and it is precisely the kind of statement that now has to be made explicit to keep this proof approach alive.

26 What These Results Do Not Yet Prove

The current Lean modules do *not* prove that the full argument fails. In particular, they do not yet address:

1. whether the masking and isolation ensemble enforces extra algebraic structure on local rules;
2. whether the switched predictors might collapse to a subclass analogous to low-depth message passing or tree computations;
3. whether the neutrality and sparsification lemmas are themselves correct in the precise form used in the paper;
4. whether a different, stronger switching theorem could bypass the present obstruction.

So the current status is best described as:

> The proof attempt is not yet broken in full, but one of its central complexity-compression transitions has not been justified, and the default combinatorics point in the opposite direction.

27 Next Attack Surface

The next natural formal target is the paper's implicit compression claim. The first and third interface steps are now isolated in Lean: `VisiblePostSwitchSurface.lean` defines the exact manuscript-visible surface $u = (z, a, b)$ together with its invariant and fork projections, and `ExactSwitchedFamily.lean` packages the exact bit-budget theorem needed to reuse the same-route ERM bound. The remaining mathematical burden is now sharper:

> After switching, the witness-bit predictor is not merely local; it is simple enough to lie in a small explicit class.

To make that rigorous, one would want a theorem of the shape:

1. define the exact visible post-switch data carried by the masked Unique-SAT ensemble;
2. define the induced witness-bit function on that data;
3. prove that this function belongs to a sharply bounded class (for example, bounded-depth circuits of size $\mathrm{polylog}(m)$, or a code family of description length $\mathrm{polylog}(m)$);
4. verify that this bound is strong enough to feed the later union-bound and compression steps.

The new module `ExactVisibleCompressionObstruction.lean` records the fallback counting barrier on that exact object: if the switched family on $u = (z, a, b)$ is still rich enough to realize *all* Boolean rules on the exact visible surface, then no s-bit code can cover it when $s < |\text{surface}|$. So any same-route rescue must now prove a genuine strict subclass theorem for the *actual* switched predictors.

If such a theorem cannot be proved, then the proof attempt should be regarded as stalled at a real mathematical gap, not a missing editorial detail.

28 Lean Excerpts

Listing 1: Core asymptotic obstruction

```
theorem polylog_isLittleO_logRadiusNeighborhood (k : Nat) {d c : Real}
    (hd : 1 < d) (hc : 0 < c) :
    polylog k =o[atTop] logRadiusNeighborhood d c

theorem logRadiusNeighborhood_not_isBigO_polylog (k : Nat) {d c : Real}
    (hd : 1 < d) (hc : 0 < c) :
    ¬ logRadiusNeighborhood d c =O[atTop] polylog k
```

Listing 2: Core counting obstruction

```
theorem no_injective_bitCode_of_lt {n s : Nat} (hs : s < 2 ^ n) :
    ¬ ∃ f : LocalRule n → BitCode s, Function.Injective f

theorem no_uniform_code_below_expInput_for_binaryLogNeighborhood
    {d m s : Nat} (hd : 2 ≤ d) (hs : s < 2 ^ m) :
    ¬ ∃ f : LocalRule (d ^ (Nat.log 2 m + 1)) → BitCode s,
      Function.Injective f
```

29 Conclusion

The present crux-first formal work yields a clean conditional diagnosis.

1. Logarithmic-radius locality is *not* the same thing as polylogarithmic complexity.

2. The full class of local Boolean rules on that visible data is far too large to fit into a polylogarithmic coding budget.

3. Therefore the program needs a separate theorem proving exceptional compressibility of the particular switched predictors it generates.

4. The v13 $P = NP$ upper-bound decoder also needs a single-message theorem for arbitrary satisfying SAT-search outputs. In Lean this is not just a slogan: correctness for all valid SAT-search outputs is equivalent to the fixed-message readout obligation, and with one satisfying witness it is equivalent to single-message constancy.

5. The v13 product-success step also needs a sequential conditional admissibility theorem; unconditional half-success marginals are insufficient, while the checked sequential conditional interface is strong enough for the two-step finite product bound.

6. A generic finite switching theorem now proves the full tower-product bound from sequential half-admissibility. The open crux is whether the manuscript's switched histories satisfy that admissibility premise on positive-mass prefixes. Empty prefixes are automatically harmless by finite-history monotonicity, while one failed nonempty prefix half-step is enough to block the tower argument. Independently, a failed final tower-product count blocks sequential admissibility and now localizes to an exact failed conditional prefix cut.

7. The v13 instantiation obligation is now fixed-field-level: both the abstract atom-ledger interface and an existential concrete-cell interface are equivalent to the generic prefix half-step. The manuscript must therefore specify the actual finite history field and prove that its fixed fibers decompose the relevant histories and satisfy the cellwise half-success bounds. The decomposition part is automatic for any finite cell map; the substantive obligation is the half-success bound inside each supplied cell, equivalently a proof that every supplied cell has at least as many next-failure prefix points as next-success prefix points. Equivalently, after the field is fixed, each successful prefix point must be injectively matched to an unsuccessful prefix point in the same field cell. This fixed-field premise is now also diagnostic in the negative direction: if the matching or prefix certificate fails, Lean extracts a supplied cell whose next-failure count is strictly smaller than its next-success count. Product-bound violations therefore localize not only to a fielded position but to a concrete bad cell in that position. This fixed-field premise is strictly stronger than global half-success: a field that reveals the successful Boolean coordinate already fails cellwise at the first step. The ordered fielded-switching interface now gives the positive bridge as well: if every supplied field certificate is proved, the tower product bound follows; if one fixed-field prefix certificate fails, the remaining fielded suffix is blocked. More generally, a positive-mass fixed-field cell that is entirely successful already blocks the certificate, so a field that exposes the next success event is admissible only in the vacuous zero-success case. Equivalently, every successful atom in a valid fixed-field certificate must be mixed with at least one unsuccessful prefix point. The same conclusion now holds for any supplied field that determines the next success bit cellwise on the current history support: on a positive-mass success prefix, such a field blocks the whole fielded suffix, and the theorem applies at any later fielded position after the preceding success-event history has been accumulated. This is strictly stronger than the global version: the Boolean-square diagonal model has a second-coordinate field that is not globally success-determining but becomes success-determining after conditioning. In the operational matching formulation, any recursive matching proof on such a positive-mass prefix directly proves that the supplied field is not success-determining on the accumulated history support. Even more concretely, every successful prefix point must have a same-cell failed prefix point; a single successful point whose supplied field cell contains only successes is already enough to rule out the matching and certificate.

8. A repeated exact-success event gives a checked positive-mass obstruction to that atom-ledger obligation: the second repeated step has conditional success 1, so no v13 switching instantiation exists for the two-step repeated Boolean-square model. It also violates the global tower-product inequality, blocking both abstract and concrete v13 certificates by the new product-bound contrapositive. The fixed-field version is now blocked at the same level: when both repeated cuts use the one-cell field, the fielded tower-product bound fails and therefore no recursive same-cell matching proof can exist. This failure is no longer only global: Lean localizes any such failure to a concrete fielded position and then to a bad supplied cell whose failures are fewer than successes. In the one-cell repeated model the second conditioned cut is explicitly the failed same-cell matching/certificate obligation, with the unique bad cell carrying failure count 0 and

success count 2.

These are currently the sharpest formally checked cruxes we have in Lean.

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